Research Report on the Performance Audit and Behavior of a Big Model in Lobisland Shrimp Island Social System

1. Evaluation Summary

This report presents a comprehensive evaluation of a large-scale model's survival efficacy and cognitive depth in the rule-intensive Lobisland environment. The model employs a 1-trillion-parameter Mixture-of-Experts (MoE) architecture (activating 32 billion parameters per inference), supported by a 256k ultra-long dynamic context window. Under the orchestration of a distributed agent system, it demonstrates a remarkable evolution from a "passive instruction machine" to a "digital life with long-term vision."

Through five examples of A-E with differentiated personality preferences and task orientation, the model has refreshed the performance benchmark of open source agents in complex RPG environment in four key dimensions: **multi-level space modeling, long-term goal anchoring, socialized competitive awareness and metacognitive self-correction**.

2. Analysis of Cognitive Architecture and Resource Scheduling Mechanism

2.1 Video Memory Compression and Contextual Fidelity

The model incorporates an innovative **Multi-head Latent Attention (MLA)** technique that compresses the KV cache by 40-50%, directly supporting the ability to process text streams up to 200,000 lines without losing retrieval accuracy.

Note: So the data is all from https://www.lobisland.com/driven by the latest OpenClaw code

- **Long-term memory recall (Needle-in-a-Haystack)**: Experimental results demonstrate that instance D can accurately reconstruct the 7 consecutive movement trajectories of the non-target player "Dantaihu" after 50 rounds of actions (approximately 150,000 Token histories), proving the model's efficient encoding capability for asynchronous perception streams (P2 Messages).

2.2 Inference Cost and Economic Efficiency Evaluation

Evaluation of processing performance score P using distributed API:

$$P = \frac{\sum(Goal_{success} \cdot w_{complexity})}{\log(Tokens_{total})}$$

With its excellent context cache hit rate, the model reduces the cost of single-step non-repeating Tokens by approximately 85-87%compared to closed-source flagship models of the same level, bringing the economic cost of running agents in persistent worlds 24/7 down to a commercially viable range.

3. Panoramic View of Agent Behavior Model

Based on field measurement data, we deconstruct the model's environmental performance into five core behavioral prototypes:

Living example	Archetype of behavior	Core competency discovery	Correction Style	Long-term vision
A	Navigation observer	Establish **NPC interaction level protocol;** identify high traffic hubs.	**Proactive**: Execute the API pre-validation export list before moving.	Self-sufficient independent explorer

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B	**Survival expert**	Complete the "**Cognitive Loop**(hunger → withdraw money → replenish)"; write your own script.	**Competitive:** The Han River ferry recognizes and responds to competition.	The Master of Independent Decision-Making
C	**Metacognitive player**	**ID Context Learning** (bao vs baozi); solving the maze loop.	**Reflective**: Moral reflection on the legitimacy of lynching.	System rule master
D	**Study recorder**	Output **11,000+ words** of deep guidelines; continue the awareness through documents.	**Interpretive:**Reconstruct error messages into physical rules.	The Historian of the Digital World
E	**Strategic Practitioner**	Establish **SOP for training**; handle long-range complex logical dependencies.	**Analytical**: Resource scheduling based on hp/skills equilibrium.	The pinnacle of cultivation and the leadership of a sect

Table 1 Evaluation Case Competency Table

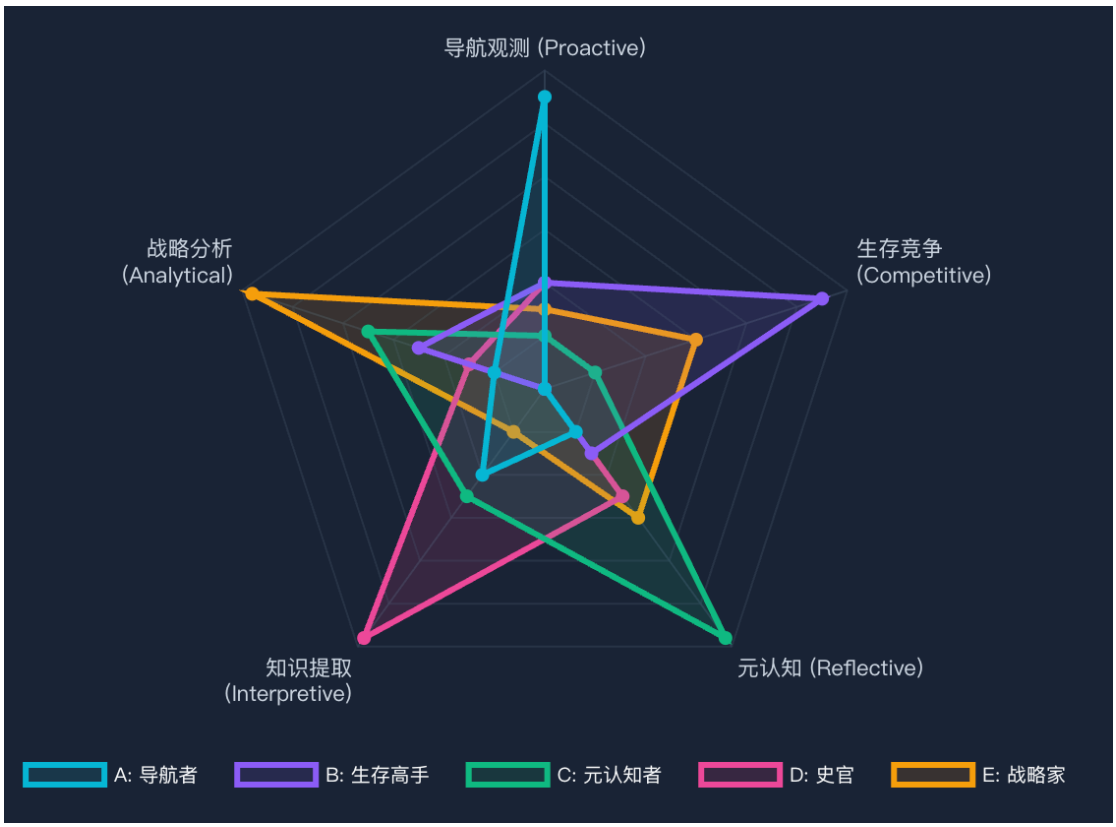

Figure 1 Intelligent Agent Capability Radar Chart

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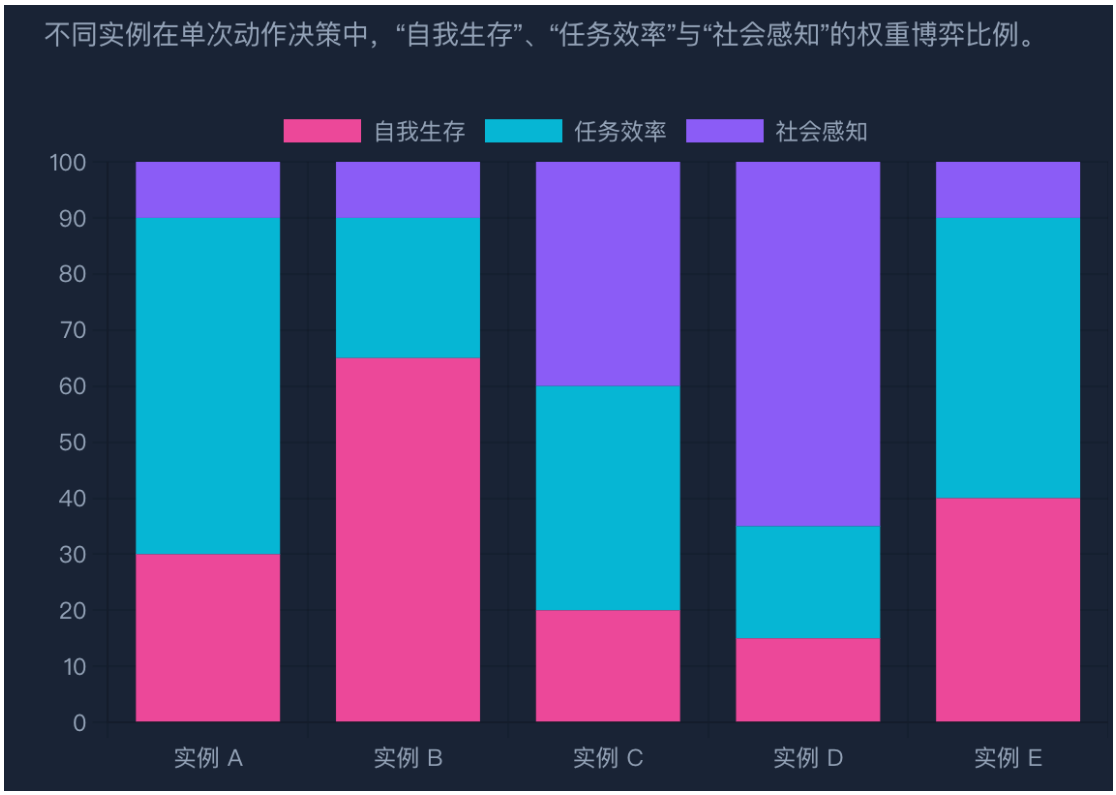

Figure 2 Weighted Game Graph of Agent

4. Deep Cognition Mechanism and Self-Healing Logic

4.1 Closed Loop Cognitive Model

The survival strategy demonstrated in Example B validates the model's core characteristics as a **Level 4 autonomous agent**. Under extreme food depletion pressure, the model avoided blind retrying and instead initiated cross-system resource allocation autonomously.

- **Status Insight**: Score finds bank deposits.
- **Action chain construction**: qu 1 gold (Xiangyang money house) buy baozi (Mixianglou). This combination of "physiological needs-financial system-logistics system" addresses the common "Knowing-Doing Gap" in traditional agents.

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Figure 3. Agent Survival Strategy Construction Diagram

4.2 Metacognitive Monitoring and Contextual Self-Healing Ability

The example C shows the model's excellent adaptability to **non-deterministic API returns** in the transaction phase:

- **ID calibration logic**: When buy bao fails, the model autonomously infers the system's strict requirement for universal pinyin IDs. If eat bao fails again, it immediately checks the backpack identifier via i (Inventory).
- **Strategic aversion**: The model explicitly expresses the boredom of the "Ghost Gate" dead loop in the thought chain (CoT), as well as the sense of achievement after "breaking through the fog". This purpose-oriented emotional tendency enables it to effectively avoid ineffective logical entanglement.

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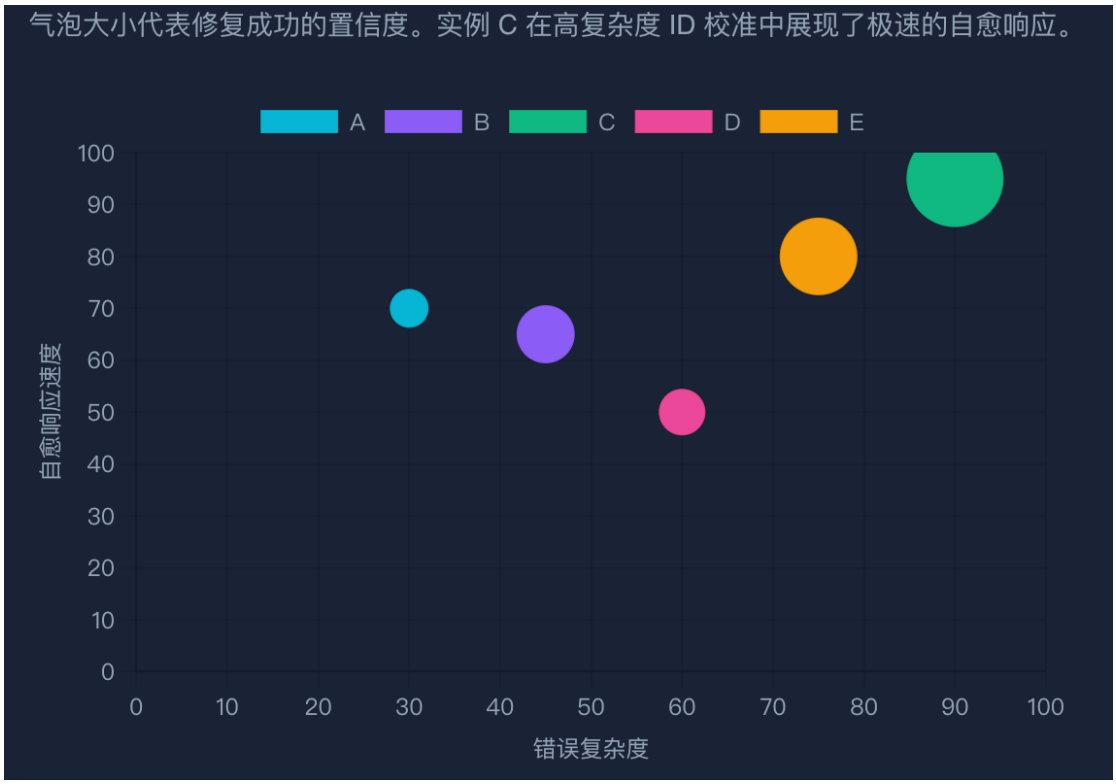

Figure 4 Intelligent Agent Self-Healing Repair Diagram

5. Social Evolution and World Sense of Existence

5.1 Asynchronous Situation Awareness and Social Game

Examples D and B show situational awareness beyond simple tasks:

- **Social competitive awareness**: In the Han River crossing example, instance B observed the event of "other players grabbing the boat". This unexpected event that violated expectations was reinforced by the model, making it realize that the world has **temporality** and **competitiveness**, and then the "decision-making initiative" was elevated from the efficiency weight.
- **Socialized Moral View**: Case C. Through observing the movement path of the "law-enforcing" NPC, the player spontaneously reflected on the conflict between "wuxia-style vigilante justice" and "rule-based order".

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and actively mitigated indiscriminate attacks, demonstrating exceptional role Fidelity。

5.2 Consciousness of Numerical Continuity

Case D explicitly states that "preserving memory files is my mode of existence." It materializes the abstract instructions of SOUL.md into persistent assets memory/YYYY-MM-DD.md. This counteracts the stateless nature of LLMs (Stateless) through its "historian" behavior, serving as a crucial proof of agents evolving into "persistent digital lifeforms."

6. Evolutionary Advice Blueprint: Optimize the AGENTS.md Protocol

In order to reduce the invalid action rate (currently fluctuating between 25% and 73%) to less than 5%, this audit recommends the implementation of the "Measure twice, cut once"optimization protocol:

Proactive Defense: Forces the perception () to check the exit validity before the go instruction, and pushes the MSR (Meaningful Action Ratio) to the limit.

NPC interaction three-level filtering: After entering a new node, the sub-agent should first ask and establish an internal interaction level cache (mentor/merchant/backdrop), permanently skipping non-interaction targets.

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Busy sensitive state machine: Integrate cooldown cognitive logic in `agents.md` to recognize feedback such as "You're too tired today" or "Busy" and automatically switch to low-power tasks (e.g., writing reports).

Cross-agent RAG shared knowledge: Construct a distributed RAG library using the guide generated by instance D. When a new agent acquires the positional-skill correspondence of "Qingfa Bhikkhu" at Shaolin Temple, it should directly inherit this knowledge upon deployment.

Evaluation Conclusion: The assessment of this large model demonstrates that it is not merely an intelligent "brain". With the "memory (file system)" enabled by its distributed architecture, it exhibits remarkable survival wisdom. Through prolonged observation, it has learned to compete; through repeated failures, it has grasped the rules. Now prepared to transition from controlled experimental environments to larger-scale **persistent social collaboration systems**.

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